AI IN SPACE AND ORBITAL OPERATIONS - VOLUME 8

The AI Space Security and Resilience Playbook

How AI defends missions across ground, link and spacecraft - and how the mission survives when defence fails - without ever mistaking an authenticated signal for a safe one.



Gates SR0-SR5 | Eight templates | 20 golden rules

An authenticated signal is not a safe mission. Space security AI is licensed per mission, per spacecraft, per link and per adversary class - never in general, and never on a penetration test passed once in a lab.

The claim ladder for this volume: mapped, assessed, hardened, exercised, transferable. A system earns the rung it can prove against a live adversary model, under exercise, on the real configuration - not the rung its audit report reached.

CHAPTER 1 - THE SPACE SECURITY DECISION BRIEF

A spacecraft cannot be recalled, patched by hand or taken offline for the weekend; security is decided before launch and lived with for the mission

Every earlier volume created attack surface while creating capability: Volume 3 automated commanding, Volume 4 created data products worth stealing and falsifying, Volume 6 built ground systems worth breaching, Volume 7 sent vehicles beyond quick help. This volume defends the whole stack - and assumes, in writing, that defence will sometimes fail. The question is never whether an adversary is interested. It is what they can reach, what that reach is worth, and whether the mission survives it.

- Name the missions, spacecraft, links and adversary classes in scope - a security claim for a hosted payload says nothing about the bus, and nothing about the ground.
- State the claim on the ladder: mapped, assessed, hardened, exercised or transferable - and the exercise evidence that would move it.
- List the standing refusals on page one: no unauthenticated commanding, ever; no single credential that can both plan and execute a critical command; no AI security tool with unlogged authority to act on the fleet.
- Price the downside in missions and trust, not in alerts: a hijacked spacecraft is a diplomatic event, a debris risk and a headline, not an IT ticket.

This volume licenses security the way Volume 1 licensed space AI: per mission, per spacecraft, per link and per adversary class, with resilience - the ability to keep a degraded mission safe - treated as a security control in its own right, not a consolation prize.

Rule: assume breach, design for survival. A mission that is only safe while nothing goes wrong is not safe.

EXECUTIVE LENS

Name the adversary classes you actually face; generic fear buys generic controls.

MISSION LENS

A spacecraft lost to intrusion is lost like any other - except someone else may be flying it.

OPERATIONS LENS

Security spend competes with margin; make it compete on exercised evidence.

Grade every stage from design to disposal, because attackers enter at the stage you graded as someone else's problem



Figure 1 - One spacecraft threaded through stacked security layers - ground, link, onboard and payload - with the lowest layer cracked in red and quarantined.

Space systems are attacked as systems. The map grades ten stages across four segments - ground, link, space and user. An ungraded stage is not a safe stage, it is an unexamined one.

Stage	Characteristic attack	First control
1. Design and supply	Compromised component, library or model	Provenance register (Chapter 8)
2. Integration and test	Test credentials and flatsat access persist into flight	Test-to-flight credential break
3. Ground systems	Ransomware, insider, cloud compromise	Segmentation (Chapter 6)
4. Command uplink	Spoofed or replayed commands	Authentication (Chapter 4)
5. Telemetry downlink	Falsified health, hidden intrusion	Integrity checks (Chapter 5)
6. Onboard software	Exploited parser, malicious update	Signed, checked updates (Chapter 7)
7. RF and navigation	Jamming, GNSS spoofing, meaconing	Multi-source navigation (Chapter 10)
8. Crosslinks	One spacecraft used to reach the fleet	Per-link trust (Chapter 11)
9. Data and users	Stolen or falsified products, account takeover	Product integrity (Chapter 12)
10. Disposal	Derelict with live credentials and propellant	Credential death at disposal (Chapter 17)

- Grade each stage per segment and per adversary class, and mark the seams - ground-to-link, bus-to-payload, operator-to-vendor - where ownership, and therefore defence, blurs.

The map is re-graded on change: a new provider, partner or copilot redraws it.

EXECUTIVE LENS

The map is the budget argument: ungraded stages are unfunded risks.

MISSION LENS

Seams are the attacker's favourite terrain; name an owner for every one.

OPERATIONS LENS

Re-grade on change, not on anniversaries.

Model who attacks, why, and what success looks like for them - a control without an adversary is decoration

Threat modelling for space starts from value, not from vulnerabilities. A constellation is worth attacking for its data, its service, its manoeuvre capability, its symbolic value and as a stepping stone to its users. Different adversaries want different things and accept different costs, so controls are licensed per adversary class, exactly as autonomy was licensed per failure mode in earlier volumes.

- Class the adversaries: state actors with patience and RF capability; organised crime after ground systems and ransom; insiders and compromised vendors; hacktivists after embarrassment; and the careless partner whose systems become someone else's entry point.
- For each class, write their success condition in one sentence - silent data theft, a public loss of control, a falsified product a customer acts on - because detection priorities follow from it.
- Model the AI-specific adversary too: the actor who poisons training data, probes the anomaly detector's blind spots, or crafts telemetry that the triage model has learned to ignore (Chapter 15).
- Refresh the model on intelligence, incidents and exercises - and record what the model got wrong last time, in the model.

A threat model that cannot be wrong cannot be improved. Grade it like any other model in this series: on its record against reality, reviewed on a cadence, with misses priced in.

Rule: if a control cannot name the adversary and the success it denies them, it is spending, not security.

EXECUTIVE LENS

Fund controls against named adversaries, not against anxiety.

MISSION LENS

The stepping-stone motive is the one boards underestimate most.

OPERATIONS LENS

Write the adversary's success condition before choosing detections.

The uplink is the mission's throat: whoever commands, flies - so every command is authenticated, authorised and checked, in that order



Figure 2 - A command beam passing through its amber authentication ring while a forged red beam dissolves against it.

Historically, many spacecraft accepted any well-formed command from anyone with a big enough dish. That era is over, and retrofitting its end is this chapter's work. Authentication proves origin, authorisation proves the origin may ask for this, and the independent checker from Volume 6 proves the command is sane for this spacecraft in this state. All three, every time, with no maintenance backdoor flying under a friendlier name.

- Authenticate every uplink cryptographically, per spacecraft, with replay protection; a command valid yesterday is evidence today, not permission.
- Separate duties in credentials: planning, approval and transmission are different keys held by different people or systems, and critical commands require two of them (Volume 6).
- Bind commands to state: manoeuvre, mode and payload commands carry preconditions the onboard system verifies before execution, so a stolen valid command still fails in the wrong state.
- Log every command, accepted and rejected, to the sacred record of Volume 6; a rejected command is an intelligence gift - someone is probing - and is reported as such, never silently dropped.

AI assists here by scoring command sequences for anomaly - unusual order, unusual hour, unusual combination - and by drafting, never by holding sole transmit authority. The system that can command the fleet is the highest-value target in the enterprise; design its compromise as a scenario, not an impossibility.

Rule: no unauthenticated commanding, no exceptions, no test mode in flight. Convenience backdoors are just doors.

EXECUTIVE LENS

The uplink is where insurance, law and engineering meet: protect it like all three.

MISSION LENS

Two-key critical commanding costs minutes; losing a spacecraft costs the mission.

OPERATIONS LENS

Rejected commands are reporting gold. Log, alert, learn.

An attacker who controls what you believe about the spacecraft controls your decisions; telemetry must be able to prove it is real

Intrusion is only half the attack. The quieter half falsifies the picture: telemetry that reports healthy batteries while a payload is exfiltrated, an orbit solution nudged until a manoeuvre is planned on false state, a detector fed a diet of almost-normal until normal itself moves. Volume 3 taught operators to distrust a green average; this chapter teaches systems to distrust a green feed.

- Sign and sequence telemetry at source where the platform allows it, and monitor for gaps, replays and impossible physics - values that are individually plausible and jointly impossible are the spoofing signature.
- Cross-check independent paths: onboard state against ground orbit determination, payload counts against power draw, reported attitude against RF and ranging observables an attacker cannot all fake at once.
- Grade AI triage on adversarial cases, not only on faults: seeded spoofing and slow-drift falsification belong in the same test library as seeded anomalies (Volume 3).
- Preserve raw telemetry unaltered (Volume 6); integrity investigations are impossible from summaries an attacker - or a helpful model - has already rewritten.

The test for this chapter is simple to state and hard to pass: could a competent adversary make the operations wall lie convincingly for one full pass? If yes, the wall is a display, not an instrument.

EXECUTIVE LENS

Belief is an attack surface. Budget for its defence explicitly.

MISSION LENS

Impossible physics is the cheapest spoof detector you own - use it.

OPERATIONS LENS

Raw records first; every summary is a claim about them.

The ground is the softest segment and the richest target: most spacecraft are attacked through the building, not through the sky

Volume 6 built trustworthy ground systems; this chapter hardens them against people. Ground stations, mission clouds, vendor remote access and operator workstations concentrate credentials, command authority and customer data in ordinary IT - which means ordinary IT attacks, ransomware and credential theft included, become space attacks the moment they land.

- Segment ruthlessly: mission operations networks separated from corporate IT and the internet, with brokered, logged crossings; a phished laptop must be an incident, not a commanding path.
- Vendor and Ground Station as a Service access is per-pass, per-mission, time-boxed and recorded, with the emergency-priority and sovereignty questions of Volume 6 asked again as security questions.
- Patch and protect the ground on an IT cadence, not a mission cadence - the ground does not get to be frozen in a secure past while attackers live in the present; the paired-configuration discipline of Volume 6 governs how, not whether.
- Exercise the loss of the ground: a mission that cannot safe its fleet from a clean backup operations centre has a single point of failure with a street address.

Insider risk is handled with the same honesty: least privilege, two-person rules for critical actions, and logs an insider cannot edit. Trust is a staffing decision; verification is a system property.

Rule: if corporate IT and commanding share a credential, a network or a fate, you do not have two systems - you have one, at the weaker one's strength.

EXECUTIVE LENS

Most space breaches will start as ordinary IT breaches. Plan the merger.

MISSION LENS

Backup operations are a security control, rehearsed, not a real-estate line.

OPERATIONS LENS

Least privilege is unpopular and undefeated; enforce it anyway.

Flight software cannot be visited, so it must verify everything it is sent - including updates, models and its own inputs

The spacecraft is a computer that parses hostile input at light-speed with no firewall team aboard. Command parsers, file uploads, model files and crosslink messages are all code-adjacent, and a parser bug at 550 kilometres is a permanent condition until a checked update survives the same path that carried the attack.

- Sign every software and model load, verify onboard before activation, and keep the tested rollback of Volume 6 as a security control: the ability to revert is the ability to evict.
- Run least capability onboard: payload processors cannot reach bus commanding, AI inference cannot write flight parameters directly, and every inter-process path is an interface with an owner and a test.
- Treat models as software with worse provenance: versioned, signed, evaluated on adversarial inputs before flight, and monitored in flight for input distributions outside their licensed envelope (Volume 2).
- Fuzz and fault-inject the parsers and loaders on the flatsat (Volume 6) with hostile files, truncated loads and replayed updates - the cases an attacker will certainly try.

Radiation and age widen every crack: memory corruption that looks like exploitation, and exploitation that hides as radiation. Forensics onboard are limited, so the design keeps the evidence - protected logs, counters and state snapshots - that Chapter 16's response will need.

EXECUTIVE LENS

Rollback is eviction capability. Test it like a weapon, keep it like one too.

MISSION LENS

A model file is an unsigned program until you sign and evaluate it.

OPERATIONS LENS

Least capability onboard is cheap at design time and priceless later.

You fly what your suppliers shipped, including their compromises; provenance is the difference between a fleet recall and a fleet mystery

Spacecraft are assembled from components, libraries, ground software, data sets and models produced by hundreds of organisations, some of which will be breached during the mission's life. The question provenance answers is forensic: when a component, library or model is implicated, which spacecraft, stations and products carry it, and what do we do on orbit about hardware we cannot swap?

- Maintain a provenance register per spacecraft: hardware lots, flight software bills of materials, model and data lineage, ground dependencies - queryable in minutes during an incident, not assembled during one.
- Verify on receipt and on load: hashes, signatures and supplier attestations checked automatically; a component that cannot prove its origin is quarantined, however urgent the schedule.
- Contract for disclosure: suppliers notify compromises affecting delivered items within a signed window, and escrow and second-source plans (Volume 6) cover the supplier who cannot ship safely for a year.
- Grade AI training data provenance with the same rigour (Volume 4): poisoned or rights-tainted data is a supply-chain defect that flies.

Provenance does not prevent compromise. It converts compromise from an unbounded mystery into a bounded, priced, actionable list - which is the entire difference between an incident and a catastrophe.

Rule: if you cannot list what is aboard within an hour of an alert, the alert owns you, not the other way round.

EXECUTIVE LENS

Provenance is incident infrastructure bought in peacetime.

MISSION LENS

Schedule pressure is the supplier attack's best accomplice; hold the quarantine.

OPERATIONS LENS

Ask suppliers how they will tell you, in the contract, before you need it.

The link is public airspace: anyone can shout, imitate or listen - resilience is designed for a contested spectrum, not a quiet one

Radio needs no network access. Uplink jamming denies commanding, downlink jamming denies knowledge, spoofing offers false versions of both, and interception is free to anyone in the footprint. Conflicts now routinely include satellite interference, so contested RF is an operating condition, not an exotic scenario.

- Design for denial: spacecraft hold safe, useful behaviour through jammed passes (Volume 3), contact plans carry alternates across stations, bands and relay paths, and no critical action depends on one contested contact succeeding.
- Detect interference as data: signal characteristics, direction and timing logged and shared, because attribution and regulatory remedy (Chapter 20) are built from operator evidence.
- Protect confidentiality by encryption on sensitive links, and authenticity by Chapter 4 - while remembering encryption does nothing against jamming; availability is a separate, physical design problem.
- Grade AI link adaptation - power, coding, routing - inside signed envelopes: an adaptive system an attacker can steer by shaping interference is a control channel handed to the adversary.

Resilience here is measured in missions continued, not decibels won. The operator who can work through interference at reduced rate, and prove what happened afterwards, beats the operator with a stronger transmitter and no record.

EXECUTIVE LENS

Contested spectrum is the normal case now; design reviews should say so.

MISSION LENS

Log interference like telemetry - future attribution depends on it.

OPERATIONS LENS

Adaptive links need envelopes, or the adversary becomes the adaptor.

A spacecraft - or a customer - that can be lied to about where it is can be steered without ever being touched

GNSS signals are faint, public and spoofable, and they underpin spacecraft navigation, ground timing and entire customer industries. A spoofed receiver does not fail; it succeeds at the wrong position and time, which corrupts orbit determination, conjunction screening (Volume 5) and every product stamped with its coordinates.

- Never navigate on one source: GNSS is fused with inertial, ranging, optical and ground determination, with source disagreement surfaced and priced into the solution's uncertainty (Volumes 2 and 7).
- Authenticate and sanity-check what can be checked: signal characteristics, clock behaviour and physical plausibility monitored continuously, with spoofing drills in the rehearsal calendar.
- Protect time as fiercely as position: ground timing derived from GNSS carries holdover and alternate sources, because a timing attack is a quiet way to desynchronise an entire operation.
- Report spoofing exposure honestly to customers whose products depend on it (Volume 4); a falsified location in a delivered product is a security incident with a customer attached.

The design principle generalises: any free, unauthenticated input the mission trusts - GNSS, public ephemerides, a partner's data feed - is an input an adversary can author. Trust is allocated per source, monitored per source, and withdrawn per source.

Rule: position and time are claims, not facts. Every claim ships with its source and its uncertainty, or it does not ship.

EXECUTIVE LENS

Single-source navigation is a standing invitation; retire it.

MISSION LENS

Timing attacks are the quiet ones - instrument for them deliberately.

OPERATIONS LENS

Customer honesty about spoofing exposure is a competitive asset, priced in trust.

A constellation that trusts itself completely is a fleet-wide incident waiting for its first foothold

Inter-satellite links turn a fleet into a network, with a network's blessings and a network's epidemiology. A compromised spacecraft, a malicious hosted payload or a partner's vehicle on a shared link standard can become a path to neighbours that were never exposed to the original attack.

- Trust per link, not per fleet: crosslink sessions authenticated per pair, scoped to the data the pair actually exchanges, and revocable per spacecraft without rekeying the constellation.
- Segment the fleet in trust zones by mission, customer and sensitivity, so a foothold in a low-sensitivity zone is a local incident, not a fleet event; zone crossings are brokered and logged like ground crossings (Chapter 6).
- Rate-limit and sanity-check crosslink behaviour: a neighbour suddenly requesting unusual data, relay volume or manoeuvre information is an anomaly the fleet SOC sees (Chapter 14), whoever owns the neighbour.
- Plan per-spacecraft isolation and credential death (Chapter 17): the fleet must be able to amputate a member cryptographically and operationally while it is still in formation.

Constellation economics push toward uniform trust; security pushes back with zones, scopes and amputation drills. The compromise is deliberate: uniformity inside a zone, suspicion between zones.

EXECUTIVE LENS

Design fleet amputation before you need it; improvising it mid-incident fails.

MISSION LENS

Partner vehicles extend your perimeter into their security posture - contract accordingly.

OPERATIONS LENS

Zones cost efficiency and buy survival; price both sides openly.

What the mission knows is worth stealing, and what it publishes is worth falsifying; protect both the vault and the voice

Earth observation archives, communications payloads and customer tasking data are the mission's commercial crown jewels, and Volume 4's products carry decisions - insurance, agriculture, disaster response - that make falsification as damaging as theft.

- Encrypt sensitive payload data in transit and at rest, with customer separation enforced cryptographically, not just by folder permissions; tasking confidentiality is itself protected, because the tasking list reveals what someone cares about.
- Sign products at creation and carry provenance through processing (Volume 4), so a customer can verify that what they received is what the sensor saw, and corrections propagate with the same signatures.
- Control the ground archive like the asset it is: access logged, exports governed, and bulk exfiltration treated as a detection case in its own right (Chapter 14).
- Classify honestly: data the mission would be embarrassed to lose, data it would be damaged to see falsified, and data whose tasking pattern is the secret - each gets controls named for its loss case.

AI raises both stakes: models can extract more from stolen archives, and synthetic imagery makes falsified products cheaper to produce. Provenance and signing stop being archival virtues and become the product's claim to be believed.

Rule: an unsigned product is a rumour with a file format. Sign at creation, verify at delivery.

EXECUTIVE LENS

Tasking lists are intelligence. Protect the questions, not only the answers.

MISSION LENS

Falsification is under-defended because theft is easier to imagine; defend both.

OPERATIONS LENS

Provenance is how real imagery survives a synthetic age.

Keys must outlive algorithms, staff and sometimes the company; a mission that cannot change its cryptography cannot change its mind about trust

Spacecraft fly for a decade or more with the cryptography they launched with, while algorithms weaken, keys leak and quantum progress reprices yesterday's mathematics. Key management is therefore a mission-design discipline: generation, storage, rotation, revocation and destruction, planned for vehicles that may be unreachable for months (Volume 7).

- Generate and hold root keys under ceremony - split knowledge, hardware protection, witnessed process - because every later assurance inherits from that room.
- Design rotation and revocation that work at orbital reality: staged rollouts by canary and wave (Volume 3), grace periods sized in passes, and a spacecraft that misses its rotation window still able to rejoin safely.
- Build crypto agility in from the start: algorithm identifiers, negotiated suites and tested migration paths, so a weakened algorithm is a planned migration, not a fleet emergency with no moves.
- Plan key death with the same care as key birth: disposal (Volume 5) includes credential and key destruction verified as far as telemetry allows, so derelicts do not fly forever with live trust.

Post-quantum migration is planned on the mission's real clock: data with a long secrecy life is exposed to harvest-now-decrypt-later today, which makes algorithm choice for archives and links a present-tense decision (Chapter 12).

EXECUTIVE LENS

Key ceremony feels theatrical until the first incident asks who held what.

MISSION LENS

Rotation sized in passes, not in calendar reminders nobody in orbit reads.

OPERATIONS LENS

Agility is bought at design time or not at all; there is no retrofit visit.

Detection spans three clocks - ground seconds, pass minutes and orbital days - and must prove itself against intrusions, not only faults

A space security operations centre watches ground networks, links and spacecraft behaviour as one picture, because attacks cross those boundaries in minutes while organisations cross them in meetings. Its core difficulty is ambiguity: an anomaly may be a fault, an environment event or an adversary, and the response differs - so detection is graded on discrimination, not just on sensitivity.

- Fuse the feeds with the raw record beneath them (Volume 6): ground security events, command and rejection logs, link interference, crosslink behaviour and onboard counters, on one timeline an analyst can actually read.
- Grade detectors on seeded intrusions in exercises (Chapter 19) as rigorously as Volume 3 graded triage on seeded faults - detection rates, time-to-detect in passes, and false-alarm load on operators all published internally.
- Protect the SOC's own AI: triage models are probed, drifted against and fed; their blind spots are red-team objectives, and their training data carries provenance (Chapter 8).
- Staff for the 3 a.m. case: alert budgets, escalation to a named security duty officer, and authority pre-delegated to isolate, revoke and safe before the meeting convenes (Chapter 16).

The SOC's product is not alerts. It is decisions with lead time: revoke before the next pass, safe before the next orbit, notify before the customer hears it elsewhere.

Rule: a detector that has never faced a seeded intrusion in exercise has an unknown detection rate, however impressive its fault record.

EXECUTIVE LENS

One timeline across ground, link and space - attacks do not respect your org chart.

MISSION LENS

Publish time-to-detect in passes; hours hide the orbital reality.

OPERATIONS LENS

Pre-delegate isolation authority. Incidents do not wait for quorum.

The same model that watches the fleet can be studied, spoofed and turned; defend the defender with the same rigour it applies to others

AI earns its place in space security: triaging security telemetry, scoring command anomaly, hunting exfiltration patterns and correlating interference across a fleet are pattern problems at machine scale. But every deployed model is also a new, soft, well-connected component whose inputs an adversary controls more often than in any earlier volume.

- License defensive AI per task class with published error rates (Volume 6): intrusion triage, command scoring, interference classification - each graded separately, each with a human decision owner for consequential actions.
- Refuse autonomous counter-action against the fleet or others: AI may isolate a ground account or recommend safing; it may not command spacecraft, revoke fleet credentials or strike back, because a manipulated defender with that authority is the attack.
- Red-team the models specifically: evasion cases, poisoned feedback, slow drift of the normal baseline and prompts crafted for operations copilots (Volume 6) are standing test categories, refreshed like any threat model.
- Monitor model inputs for out-of-distribution attack the way Volume 7 monitored environments: a detector operating outside its exercised envelope demotes to advisory, loudly.

The pattern is the series' pattern: capability inside envelopes, demotion on triggers, humans owning consequence. Security AI that exempts itself from the discipline it enforces will eventually enforce nothing.

EXECUTIVE LENS

The defender is part of the attack surface. Put it on the map, Chapter 2.

MISSION LENS

No autonomous counter-action: manipulated authority is the adversary's prize.

OPERATIONS LENS

Demote detectors outside their exercised envelope, in the open.

You cannot seize the server, pull the plug or fly out a disk; response is conducted by radio, in passes, against an adversary who may be listening

Orbital incident response runs on the pass clock with partial knowledge. Containment means revoking, isolating and safing through the same links the adversary may be watching or contesting, and eradication may be impossible until a signed update can be built, checked and flown - the Volume 6 pipeline running as an emergency service.

- Pre-write the playbooks per scenario class - ground compromise, suspected onboard intrusion, spoofed commanding, falsified products - with authorities, customer and regulator notifications, and the order of revocation rehearsed before it is needed.
- Preserve evidence first where physics allows: protected logs and counters captured, raw telemetry sealed (Volume 6), because the post-incident argument with insurers, customers and states runs on records.
- Safe in a way that denies the adversary value: isolation postures that protect propulsion, power and credentials first, accept payload loss before bus loss, and never let response actions create a conjunction or debris risk (Volume 5).
- Decide disclosure on the pre-signed plan (Chapter 20), not under pressure: who is told, in what order, with what verified facts, while facts are still arriving in passes.

Recovery is a re-licensing event, not a return to normal: the affected systems earn trust back through evidence at a gate (SR4), exactly as autonomy did in earlier volumes. Normal is a claim, and claims need proof.

Rule: rehearse response on the pass clock, with the adversary played live. A playbook never exercised is a hypothesis with a title page.

EXECUTIVE LENS

Response authority pre-delegated, in writing, per scenario class.

MISSION LENS

Payload loss before bus loss; bus loss before debris risk. In that order.

OPERATIONS LENS

Recovery is a gate, not a mood. Evidence re-earns the licence.

Resilience is the mission continuing safely at reduced value - designed degradations, rehearsed, with nothing important improvised under attack

Hardening fails somewhere, sometime. Resilience is what the mission owns next: pre-designed degraded modes that preserve safety, custody and the most valuable functions, shed the rest in a signed order, and keep working through jamming, compromise or the loss of ground, links or fleet members.

- Design the degradation ladder per mission: full service, reduced service, custody-only, safe-and-silent - each with entry triggers, what is shed, what is protected and what exit back upward requires.
- Keep the safety core independent and simple, as with FDIR throughout this series: deterministic safing that does not depend on the networks, models or credentials under suspicion.
- Kill credentials at end of life and on amputation: disposal and isolation include verified credential death (Chapter 13), so yesterday's spacecraft cannot be tomorrow's foothold or weapon.
- Exercise degradation like a product feature: timed transitions in rehearsals, customer communication templates ready, and degradation time counted honestly in the economics of Chapter 21.

Graceful degradation is also a negotiating asset: customers and insurers pay for missions that bend in writing rather than break in public. The ladder, exercised, is a sales document with evidence behind it.

EXECUTIVE LENS

Degradation ladders are designed in peacetime or improvised in wartime. Choose.

MISSION LENS

The safety core trusts nothing that is currently under suspicion.

OPERATIONS LENS

Credential death is a disposal duty, verified, not assumed.

Not every attack arrives as data: rendezvous, inspection and debris are physical sentences in the same security grammar

Co-orbital vehicles can inspect, shadow, interfere or - at the limit - collide. Ground lasers can dazzle sensors. A security programme that models only cyber paths will be surprised by physics, and a response that manoeuvres carelessly can create the conjunction Volume 5 exists to prevent.

- Treat uncooperative proximity as a security event with a playbook: detection and characterisation from space surveillance data, notification thresholds, manoeuvre options pre-planned with secondary screening (Volume 5), and evidence preserved for Chapter 20's attribution and diplomacy.
- Protect sensors and payloads by design where feasible: shutters, filters and safe pointing modes for dazzling, with autonomous protective responses inside signed envelopes that cannot be tricked into self-denial by a passing flash.
- Harden the obvious physical dependencies: ground sites, fibre paths, power and the small number of people whose absence or coercion would matter - the unglamorous perimeter where real operations live.
- Coordinate with orbital safety, always: security manoeuvres enter the same decision authority as conjunction manoeuvres (Volume 5), because a panicked evasion that creates debris is a self-inflicted counterspace act.

Physical threats also discipline the rhetoric: most proximity is lawful, much inspection is ambiguous, and the playbook's job is proportionate, evidenced response - not escalation with a propulsion budget.

Rule: one authority decides manoeuvres, whatever provoked them. Security urgency never bypasses secondary screening.

EXECUTIVE LENS

Characterise before you manoeuvre; evidence outlives the encounter.

MISSION LENS

Protective autonomy needs envelopes, or a flashbulb becomes a denial weapon.

OPERATIONS LENS

Physical perimeters are security controls. Walk them, literally.

The only honest detection rate is the one a live adversary produced; exercise the whole system, including the people and the paperwork

Penetration tests scoped to a server prove a server. Space security exercises play the mission end to end: a red team with an adversary class brief (Chapter 3), freedom across ground, link simulation and flatsat, and a clock that runs in passes - while the blue side works with real staffing, real escalation and the real temptation to explain anomalies away.

- Scope exercises to mission outcomes, not components: can the red team command, falsify, steal or deny - and at which rung of Chapter 2's map did defence actually engage?
- Include the AI in the target set and in the defence (Chapter 15): evading the triage model, poisoning its feedback and spoofing the SOC's picture are scored objectives, not unfair tricks.
- Run tabletop exercises for the decisions technology cannot make: disclosure, ransom, attribution claims, customer notification - timed, minuted, with executives playing their real roles.
- Close findings like flight findings: owners, dates, re-test evidence, and a register the SR gates read. A finding repeated at the next exercise is a governance failure, named as such.

Exercises are also where the halt culture is built: a junior analyst who safes a spacecraft on a suspicion that proves wrong must be visibly backed, or the next suspicion will wait for certainty that arrives one pass too late.

EXECUTIVE LENS

Score exercises on mission outcomes. Component passes comfort nobody under attack.

MISSION LENS

Executives play themselves in tabletops; delegation is the finding everyone expects.

OPERATIONS LENS

Back the false-alarm safer publicly, or lose the next true alarm.

In space, a cyber incident is also a state problem: responsibility, attribution and escalation run through governments, not just counsel

The Outer Space Treaty makes states responsible for national space activities and liable for damage; national licensing (Volume 1) increasingly carries cybersecurity conditions; interference with space systems engages spectrum regulation and, at scale, international politics. An operator's incident is therefore never only an operator's incident.

- Know the notification map before it is needed: licensing authority, spectrum regulator, insurer, affected customers, partner states where their assets or data are touched - with thresholds and clocks taken from the actual licences and contracts, not assumed.
- Grade attribution claims like navigation solutions (Chapter 10): source, confidence and uncertainty stated, because public attribution is an escalatory act and a wrong one is unrecoverable.
- Preserve the evidence chain to a standard others can rely on (Chapters 5, 14 and 16): shared interference and intrusion data is how norms, insurance and collective defence get built in this domain.
- Practise the norms you want to live under: no testing against others' spacecraft, responsible disclosure of vulnerabilities affecting shared systems, and coordination data shared in the formats the community can use (Volume 5).

Disclosure handled early and factually is consistently cheaper - in fines, in trust and in diplomacy - than disclosure extracted later. The playbook writes the order and the clocks in advance because neither can be invented credibly mid-incident.

Rule: attribution is a claim with an uncertainty attached. Publish it that way, or do not publish it.

EXECUTIVE LENS

Notification clocks live in licences and contracts; extract them in peacetime.

MISSION LENS

Shared evidence builds the norms that protect everyone, including you.

OPERATIONS LENS

Early factual disclosure is the cheapest version of every incident.

Security is bought with the mission's money, so count what it buys: days the fleet operated safely under real threat conditions, fully loaded

Security proposals arrive as fear or as compliance; this series counts in verified outcomes. The unit here is the verified resilient satellite-day: a day the spacecraft operated within its licensed security posture - controls exercised, detections proven, degradation ladder ready - divided into the fully loaded cost of achieving it.

- Fully loaded means fully loaded: tooling, SOC staffing, exercises and red teams, key ceremonies, supplier assurance, incident costs and the engineering time security reviews consume - the costs proposals most often leave in other budgets.
- Claim savings only at the rung exercised: automation that a red team bypassed has not released analyst hours, and a control never tested is a cost with a hope attached (Chapter 19).
- Price degradation honestly: days spent in reduced modes count at their verified value, which keeps resilience improvements - faster recovery, shallower degradation - visible as economic gains, not just engineering pride.
- Compare like with like: cost per verified resilient satellite-day against the fleet's own baseline over time, with incident losses shown beside it, never hidden inside it.

The number disciplines both directions: it stops security buying theatre, and it stops finance stripping controls whose exercised value exceeds their cost by the price of one lost spacecraft.

EXECUTIVE LENS

Count security like operations: verified days, fully loaded cost.

MISSION LENS

One lost spacecraft reprices every control you declined. Keep that arithmetic visible.

OPERATIONS LENS

Recovery speed is economic value; measure degradation in days and money.

Security is a named authority, a register and a rehearsed halt - or it is a poster in the operations corridor

The operating model gathers the volume: a chief space security authority with a seat beside the mission and operations leads, a live security register per mission, and gates SR0-SR5 at which claims are earned, reduced or refused on evidence.

- Roles: space security authority, mission security lead per programme, SOC duty officer, key custodians and a supplier assurance owner - named people, with deputies, and authority that survives their holidays.
- Every licence states its rung, scope per spacecraft and link, adversary classes covered, exercise evidence, expiry and demotion triggers; demotion is automatic on triggers - a confirmed bypass, an unclosed critical finding, an expired exercise - and promotion is earned back at a gate.
- Anyone can halt: an operator, analyst or engineer who believes commanding, a link or a product is compromised can safe the affected scope without prior approval, and the halt is reviewed for its reasoning, never punished for its caution.
- Report to the board in the ladder's language: posture by rung and fleet percentage, time-to-detect in passes, exercise results and the register's oldest unclosed findings - one page, no theatre.

Space systems ask the public to trust machines no one can visit, carrying services everyone uses. This operating model exists so that trust is a register, an exercise record and a rehearsed halt - evidence an adversary has already tested, rather than assurance they have not yet bothered to.

EXECUTIVE LENS

Automatic demotion, gated promotion - the ratchet turns toward evidence only.

MISSION LENS

The halt belongs to everyone; its review belongs to the record.

OPERATIONS LENS

Board reporting in rungs and passes keeps security honest at the top.

WORKED EXAMPLE - AURORA ORBITAL, PART 1

The exercise: a red team, a ground breach and a fleet that had graded itself too generously



Figure 3 - The Aurora security operations wall: the fleet in cyan, one spacecraft quarantined in red under its amber containment dome.

Fictional worked example. All organisations, figures and events are illustrative.

Aurora Orbital (Volumes 1-7) commissioned its first full-mission security exercise before scaling its constellation. The vendor claimed a 51 percent security-operations saving and 57,000 analyst hours per year, with the fleet self-graded at rung 5, transferable, on the strength of a component penetration test.

What the exercise and audit found

- Verified saving was 16 percent: 24,600 hours verifiably released and 14,800 banked, once SOC staffing, exercises, key management and supplier assurance were costed back in.
- The red team entered through a vendor remote-access account whose per-pass scoping had lapsed into permanent access, reached the planning network, and was stopped only by the two-key rule at commanding (Chapter 4) - the control that had been mocked as ceremony.
- Seeded spoofed telemetry reporting healthy batteries on a degrading spacecraft ran for six passes before an impossible-physics check (Chapter 5), not the AI triage model, caught it; the model had been drifted against during the exercise's first week.
- The provenance register (Chapter 8) answered the implicated library query in 40 minutes - and found it on two spacecraft and one ground system nobody had listed.

None of this required a real adversary. It required the map to be believed, a red team allowed to win, and a register that could be queried in minutes.

EXECUTIVE LENS

A self-grade is a claim; the exercise is the audit.

MISSION LENS

The mocked two-key ceremony was the control that held.

OPERATIONS LENS

Six passes of spoofed health is the finding - price it, fix it, re-test it.

WORKED EXAMPLE - AURORA ORBITAL, PART 2

SR5 granted in part: the posture that survived was smaller and worth more

Fictional worked example, continued. All figures illustrative.

Function	Claimed	Licensed at SR4
Command authentication and two-key control	Rung 5	Rung 4 - exercised, held under red-team attack
Ground segmentation and vendor access	Rung 5	Rung 3 - assessed and hardened, permanent access revoked, re-exercise due
Telemetry integrity monitoring	Rung 4	Rung 3 - physics checks live, AI triage demoted to advisory after drift finding
Crosslink trust zones	Rung 4	Rung 4 - zones and per-link revocation exercised, amputation drill passed
Autonomous counter-action by security AI	Standard feature	Refused - AI recommends, humans command (Chapter 15)
Fleet-wide single trust domain	Rung 5	Refused - replaced by trust zones (Chapter 11)

Cost per verified resilient satellite-day settled at 0.85 against a 0.98 pre-programme baseline, fully loaded, with exercise and incident-drill time inside the number. SR5 scale was granted in part: the posture extends to new spacecraft one zone at a time, each earning its rung under exercise.

The vendor relationship improved under the new terms - per-pass access, exercise rights, published detection rates - because the contract finally described the product Aurora actually needed: not an unbreachable fleet, but a fleet whose breaches are detected in passes, contained by design and survived.

EXECUTIVE LENS

Scale in part is the normal, healthy outcome of an honest gate.

MISSION LENS

Demoting the drifted triage model protected every decision downstream of it.

OPERATIONS LENS

A smaller true posture compounds; a grand false one detonates in public.

THE SPACE SECURITY STANDARD

One page, to be read before any spacecraft is declared secure

The Security Standard. We license space security per mission, per spacecraft, per link and per adversary class. No unauthenticated commanding, ever. Critical commands take two keys. Raw records are never altered. Defensive AI recommends; humans command. Anyone can halt a compromised scope, and recovery is earned back at a gate on evidence.

- We claim only the rung we exercised: mapped, assessed, hardened, exercised, transferable.
- We assume breach and design degradation ladders that keep the mission safe at reduced value.
- We keep provenance queryable in minutes and credentials killable at disposal.
- We grade detection on seeded intrusions, in passes, and publish the rates internally.
- We count cost per verified resilient satellite-day, with exercises and incidents inside the number.

Print it. Sign it per mission. A standard that lives only in this playbook governs nothing.

EXECUTIVE LENS

Sign it per mission; an unsigned standard is a poster.

MISSION LENS

The standard is short because the pass is short.

OPERATIONS LENS

Hold the standard at launch and scaling gates, where it still costs only schedule.

SIX FAILURE PATTERNS

How space security programmes actually go wrong

Pattern	What happens	Antidote
1. The component pass	A server penetration test is read as a mission security grade	Exercise mission outcomes end to end (Chapter 19)
2. The permanent guest	Vendor and test access granted for a day persists for the mission	Per-pass scoping, automatic expiry, access audits
3. The trusted fleet	One foothold becomes a constellation through uniform crosslink trust	Trust zones, per-link revocation, amputation drills
4. The lying wall	Spoofed telemetry keeps operations confident while the spacecraft is lost	Impossible-physics checks, independent observables (Chapter 5)
5. The armed defender	Security AI with command authority is manipulated into acting for the attacker	AI recommends, humans command; no autonomous counter-action
6. The immortal derelict	A disposed spacecraft flies on with live credentials and propellant	Verified credential death at disposal (Chapter 17)

Every pattern was visible in an exercise before it was visible in an incident. The patterns repeat because each begins as a reasonable convenience under schedule pressure, granted on the ground, paid for in orbit.

EXECUTIVE LENS

Patterns are cheaper to recognise in this table than in an incident report.

MISSION LENS

Convenience is the common ancestor; govern it, not just its children.

OPERATIONS LENS

Exercises are where patterns go to be caught cheaply.

30 QUESTIONS

Asked at every gate, answered with evidence or not at all

- Which missions, spacecraft, links and adversary classes does this licence cover?
- What rung is claimed, and what live exercise proves it?
- What is each adversary class's success condition, in one sentence?
- Can any single credential plan and execute a critical command?
- Where does unauthenticated commanding still exist, including test modes?
- Could spoofed telemetry keep the wall green for a full pass?
- What shares a fate with corporate IT - credentials, networks, people?
- Which supplier compromise would hurt most, and how fast would we know?
- What is aboard each spacecraft, queryable in how many minutes?
- How does the mission operate through a jammed or spoofed pass?
- Which navigation and timing sources are trusted, and what disagrees with them?
- Can one spacecraft be amputated from fleet trust, and when was it last drilled?
- Are products signed at creation, and can customers verify them?
- When were keys last rotated, and what happens to a spacecraft that misses its window?
- What is the time-to-detect, in passes, against seeded intrusions?
- Has the defensive AI been red-teamed for evasion, drift and poisoning?
- Who can isolate, revoke and safe at 3 a.m., in writing?
- Are response playbooks written per scenario, with notification clocks extracted?
- What is the degradation ladder, and when was each rung last entered in exercise?
- Is credential death verified at disposal, per spacecraft?
- Who decides security manoeuvres, and do they pass secondary screening?
- What did the last red team achieve, and which findings remain open?
- Are attribution claims graded with sources and uncertainty?
- What is the fully loaded cost per verified resilient satellite-day?
- Which savings depend on thinning exercises, staffing or provenance?
- What expires - keys, access, exercises, licences - and what does expiry demote?
- Which seams between segments have no named owner?
- How are rejected commands and interference events reported and shared?
- What evidence would make us reduce this licence tomorrow?
- Who halted something last, and were they backed in the record?

EXECUTIVE LENS

Thirty questions, one standard: evidence or it did not happen.

MISSION LENS

Ask them at scaling gates, not after the first real incident.

OPERATIONS LENS

Any unanswered question is a rung the licence has not earned.

Six gates, each with the evidence that opens it

Gate	Meaning	Evidence required
SR0 - Frame	Scope the security mission	Security map drafted per segment; adversary classes named; standing refusals signed
SR1 - Map	Attack surface known	Ten-stage map graded; provenance register live; seams owned
SR2 - Assess	Weaknesses found honestly	Threat models per class; assessments of ground, link, onboard and data; key plan set
SR3 - Harden	Controls built and configured	Authentication and two-key live; segmentation enforced; trust zones and degradation ladder built
SR4 - License	Posture relied on at earned rungs	Full-mission exercise passed; detection rates published; halt honoured in drill
SR5 - Scale	Extend missions and fleet	Per-zone exercise evidence for new scope; unit cost verified with exercises beside it

First 90 days

- **Days 1-30.** Build the security map and provenance register; sign the standing refusals; revoke every permanent vendor and test access.
- **Days 31-60.** Enforce authentication, two-key commanding and segmentation; design trust zones and the degradation ladder; extract every notification clock.
- **Days 61-90.** Run the first full-mission exercise with a live red team, seeded spoofing and a 3 a.m. halt; hold the first SR4 gate on evidence.

EXECUTIVE LENS

Gates open on exercises, not on audit reports.

MISSION LENS

The 90 days end with a shorter claim and a longer register.

OPERATIONS LENS

SR5 adds scope one zone at a time; it never widens a licence by itself.

What an auditor, a regulator, an insurer or an inquiry will eventually ask for

The evidence pack, per mission

- Security map history: grades per stage and segment, with re-grades on change.
- Threat models per adversary class, with their record against reality.
- Command and rejection logs, access records and key ceremony and rotation records.
- Provenance register extracts per spacecraft, ground system and model.
- Exercise and red-team reports with findings closure and re-test evidence.
- Detection scoring: seeded intrusion results, time-to-detect in passes, false-alarm load.
- Incident and degradation records, including halts, revocations and recovery gates.
- Interference and spoofing logs shared with regulators and community schemes.
- Disposal records showing verified credential and key death.

Sector annex - where the emphasis moves

Sector	Emphasis
Communications constellations	Ground and user-segment scale, crosslink zones, service denial economics
Earth observation	Product falsification, tasking confidentiality, archive exfiltration
Government and defence missions	State adversary classes, attribution discipline, allied data sharing
Crewed and lunar missions	Life-safety commanding, proximity playbooks, long-silence key management
Small-satellite and hosted payloads	Bus-payload separation, shared-ground risk, proportionate controls
Navigation and timing services	Spoofing at customer scale, holdover, honest exposure reporting

EXECUTIVE LENS

Assemble the pack before anyone asks; afterwards is too late and it shows.

MISSION LENS

The annex moves emphasis, never the standard.

OPERATIONS LENS

The exercise record is the evidence pack's spine.

EIGHT FILL-IN TEMPLATES

The licence, made paperwork

Template 1 - Security Licence Card
Mission: ___ | Spacecraft / link: ___ | Rung: ___ | Adversary classes: ___ | Expiry: ___ | Authority: ___

Template 2 - Security Map Grading Sheet
Stage / segment: ___ | Grade: ___ | Characteristic attack: ___ | First control: ___ | Owner: ___

Template 3 - Adversary Class Brief
Class: ___ | Motive: ___ | Success condition: ___ | Capabilities: ___ | Controls that deny them: ___

Template 4 - Command Authority Record
Command class: ___ | Keys required: ___ | Holders: ___ | State preconditions: ___ | Checker: ___

Template 5 - Exercise and Red-Team Record
Scenario: ___ | Adversary class played: ___ | Achieved: ___ | Detected in (passes): ___ | Findings / owners: ___

Template 6 - Incident and Revocation Log
Incident: ___ | Scope isolated: ___ | Credentials revoked: ___ | Notifications / clocks: ___ | Recovery gate: ___

Template 7 - Degradation Ladder Card
Mission: ___ | Rungs and triggers: ___ | Shed order: ___ | Protected core: ___ | Exit evidence: ___

Template 8 - Verified Resilient Day Workbook
Period: ___ | Fully loaded security cost: ___ | Verified resilient satellite-days: ___ | Cost per day: ___ | Claimed / released / banked: ___

Maps, briefs, authorities, exercises, revocations, ladders and cost - each short enough to complete during a pass, which is when several of them will be used. Paperwork from logs is evidence; from memory, fiction.

The volume in twenty lines, and the series in ten

- License per mission, spacecraft, link and adversary class.
- An authenticated signal is not a safe mission.
- Assume breach; design for survival at reduced value.
- No unauthenticated commanding, ever, in any mode.
- Critical commands take two keys and an independent check.
- Position, time and telemetry are claims - ship their uncertainty.
- The ground is part of the spacecraft's security, at IT speed.
- Provenance queryable in minutes, or the alert owns you.
- Trust per link and per zone, never per fleet by default.
- Sign products at creation; unsigned output is a rumour.
- Keys rotate in passes; credentials die at disposal.
- Detection is graded on seeded intrusions, in passes.
- AI recommends; humans command. No autonomous counter-action.
- Response runs on the pass clock, rehearsed with a live adversary.
- Degradation ladders are designed, exercised and priced.
- One authority decides manoeuvres, whatever provoked them.
- Attribution ships with its uncertainty, or not at all.
- Cost per verified resilient satellite-day, exercises inside.
- Anyone can halt a compromised scope, and is backed for it.
- A smaller true posture beats a grand false one.

AI in Space and Orbital Operations - the series

Vol	Title	Gates	Status
1	Space Strategy	SS0-SS5	Built
2	Mission Design and Autonomy	MD0-MD5	Built
3	Satellite and Constellation Operations	CO0-CO5	Built
4	Earth Observation and Data	EO0-EO5	Built
5	Conjunction, Debris and Orbital Safety	CD0-CD5	Built
6	Ground Systems and Mission Software	GS0-GS5	Built
7	Lunar and Deep-Space Operations	LD0-LD5	Built
8	Space Security and Resilience	SR0-SR5	Built - this volume
9	Space Economics and Fleet Finance	EF0-EF5	Next
10	Space Capstone and Assurance	SC0-SC5	Planned

Volume 9, Space Economics and Fleet Finance, is next. The series closes at ten with the Space Capstone and Assurance.